Mohammad Umar Haris

Data Scientist – Generative AI & Machine Learning

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Professional Summary

Data Scientist with 4+ years of experience in analytics, machine learning, and Generative AI application development across insurance, logistics, and pharmaceutical domains. Hands-on in designing and deploying LLM-based pipelines, predictive models, and cloud-based ML infrastructure using Python, Azure, Databricks, MLflow, and FastAPI. MSc Data Science (Distinction) with applied expertise in NLP, forecasting, model deployment, and CI/CD automation via GitHub Actions.

Core Skills

- **Generative AI & LLMs:** LLM API Integration, Prompt Engineering, Azure OpenAI, Text Summarisation Pipelines, Copilot Integration
- **ML & Statistics:** Predictive Modelling, Classification, Regression, Time Series Forecasting, NLP, Feature Engineering, A/B Testing, Hypothesis Testing, Model Evaluation
- **MLOps & Deployment:** MLflow, FastAPI, GitHub Actions (CI/CD), Azure ML, Docker, Model Versioning, REST API Deployment
- **Programming & Analysis:** Python (Pandas, NumPy, Scikit-learn), SQL, R
- **Cloud & Data Engineering:** Azure Databricks, Azure Data Factory, Azure SQL, ETL Pipelines
- **BI & Visualisation:** Power BI, Tableau, Google Data Studio
- **Practices:** Agile (Scrum), Stakeholder Management, Cross-functional Collaboration, Model Validation

Professional Experience

Data Scientist **Sept 2023 – Jan 2026**

UK

- Built and deployed an end-to-end Generative AI pipeline that transcribed call recordings via Azure Speech-to-Text and used Azure OpenAI (GPT-4) for summarisation, intent and sentiment extraction, and automated recovery-action generation.
- Integrated a Microsoft Copilot chatbot into agent CRM workflows, surfacing LLM-generated insights directly within case-recovery tools.
- Served ML models and LLM pipeline outputs as REST APIs using FastAPI, with MLflow for experiment tracking, model versioning, and the model registry.
- Automated testing, validation, and deployment through GitHub Actions CI/CD on Azure infrastructure.
- Developed Azure Databricks ETL pipelines and Power BI dashboards for cost, utilisation, and performance tracking — improving planning accuracy by 22% and reducing reporting turnaround by 35%.

Junior Data Scientist (Part-time) **Oct 2022 – Aug 2023**

System Engineering (AI & Data Consultancy), London, UK

- Contributed to a classification-based product recommendation engine, framing recommendations as a binary purchase-probability problem to support more personalised targeting.
- Built and evaluated classical ML models in scikit-learn (Logistic Regression, Random Forest, XGBoost), supporting a baseline that improved on the legacy rule-based approach.
- Engineered behavioural and transactional features from SQL data using Pandas and NumPy, handling missing-value imputation, skewed distributions, and categorical encoding.
- Supported model evaluation under class imbalance using SMOTE and class weighting, tracking ROC-AUC and F1-score to assess model quality.
- Assisted the product team with A/B test analysis to measure the impact of ML-based recommendations on engagement metrics such as click-through rate.

Associate Analyst **Dec 2021 – Sept 2022**
Cognizant Technology Solutions, India

- Engineered data quality and validation pipelines for Google Ads content policy, processing 7,200+ monthly records at 99% accuracy and systematically resolving 100+ weekly extraction anomalies — achieving top 5 data accuracy performance across 30+ teams.
- Automated policy validation workflows for a financial services client using rule-based logic and structured data checks, eliminating manual review overhead and implementing process controls that reduced data discrepancy rates by 30%.
- Designed and deployed KPI dashboards in Google Data Studio and Google Sheets with automated data feeds, enabling real-time governance reporting and performance tracking across multiple business units.
- Executed A/B testing frameworks on operational workflows driving 15–20% conversion uplift; developed an Excel-based decision modelling tool with structured logic that reduced policy evaluation time by 30%, standardising a previously inconsistent manual process.

Data Analyst (Part-time) **Jan 2020 – Dec 2021**
Inletware Ltd

- Performed end-to-end analysis of pharmaceutical e-commerce data using SQL, cleaning and transforming raw transactional and inventory datasets to support demand planning and operational reporting.
- Built time-series forecasting models to anticipate product demand and reduce stockouts, helping the team plan inventory more accurately across key product categories.
- Designed and maintained 30+ Power BI and Tableau dashboards tracking sales, inventory, and fulfilment metrics, improving reporting efficiency by ~20% and giving stakeholders self-serve visibility into KPIs.
- Consolidated data from Azure SQL and internal sources into clean, analysis-ready tables, and automated recurring reports to replace manual spreadsheet preparation.

Selected ML / Data Science Projects

- **Generative AI Call Analytics Pipeline** *(LLM APIs, Azure, FastAPI, MLflow, GitHub Actions, speech-to-text)* — Production GenAI pipeline integrating transcription, LLM-based summarisation, sentiment-driven analysis, and automated action generation; served via FastAPI on Azure with MLflow tracking and CI/CD.
- **Anticancer Molecule Activity Prediction** *(Python, Scikit-learn, RDKit, ensemble methods)* — Classification pipeline predicting anticancer activity from molecular descriptors and fingerprints, using feature selection and cross-validation.
- **Gout Disease NLP Classification** *(Python, NLP, TF-IDF, Scikit-learn)* — NLP pipeline classifying gout-related clinical text using TF-IDF vectorisation and supervised models with text preprocessing and evaluation.
- **Demand Forecasting & Inventory Optimisation** *(Python, time-series models, SQL, Power BI)* — Forecasting models on pharmaceutical e-commerce data to prevent stockouts and improve planning accuracy.

Education

MSc Data Science and its Applications — Distinction 2022 – 2023
University of Essex, UK

Bachelor of Pharmacy (B.Pharm) — First Class 2017 – 2021
Jamia Hamdard University, India